DATA MINING RESEARCH REPORT

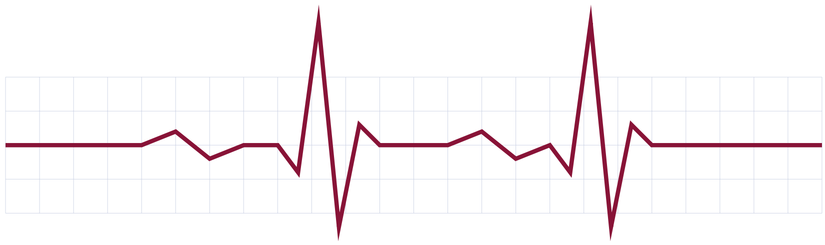
Department of Computer Science & Information Systems

DATA MINING & PREDICTIVE ANALYTICS PROJECT

HEART DISEASE RISK PREDICTION

USING DATA MINING AND MACHINE LEARNING

A Rigorous Scientific Study, Hypothesis Testing Suite, 7-Tier Risk Stratification & Interactive Clinical Decision Support System



Academic Year:	2025–2026
Course Module:	Data Mining & Knowledge Discovery (`CS-504`)
Simulated Contributors:	`AISD05` (Data Audit, EDA & Hypothesis Suite)
	`SIAD19` (Preprocessing Pipeline & ML Benchmark)
	`RSD20` (Streamlit UI, Predictor App & LaTeX Report)
Technology Stack:	Python • Pandas • Scikit-Learn • SciPy • Streamlit • GitHub • LaTeX

DEMONSTRATION / REFERENCE PROJECT NOTICE

This scientific report is an educational reference project created to demonstrate technical depth, statistical rigor, Git workflow, and presentation of a Data Mining study. A single GitHub account was used to publish the repository with simulated contributor branches (`AISD05`, `SIAD19`, `RSD20`).

Copyright & Project Declaration

Abstract

Cardiovascular diseases (CVDs) represent the leading cause of global mortality, demanding accurate, early, non-invasive risk identification. This scientific report documents a complete, publication-grade Data Mining investigation on the official Kaggle dataset `uditjain13/heart-disease-risk-2026`. We execute a rigorous data quality audit ($N = 9,000$ patient records, 25 clinical attributes), leakage-free preprocessing via scikit-learn `ColumnTransformer` pipelines, exploratory data analysis, and a suite of six non-parametric statistical hypothesis tests ($\alpha = 0.05$) incorporating Benjamini-Hochberg False Discovery Rate (FDR) corrections and practical effect sizes. We benchmark six supervised classification algorithms: Zero-R Baseline, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Naive Bayes, and Support Vector Machine (SVM), using 5-fold Stratified Cross-Validation and hyperparameter grid search. The Support Vector Machine (RBF kernel) achieved leading diagnostic performance with an Accuracy of 89.17%, Weighted F1-score of 0.8898, and ROC-AUC of 0.9422. Multi-perspective feature importance consensus identified Maximum Heart Rate Achieved, ST Segment Depression, Age, and LDL Cholesterol as the primary clinical risk drivers. Furthermore, we elaborate on the healthcare application of clinical decision support systems (CDSS), wearable device monitoring integration, and a 7-tier clinical risk stratification framework (0% to 100%) for triage and resource allocation.

Keywords: Data Mining; Heart Disease Risk Prediction; Supervised Classification; Support Vector Machines; Statistical Hypothesis Testing; Benjamini-Hochberg FDR; Feature Importance; Clinical Decision Support Systems; Streamlit.

Contents

List of Figures

List of Tables

List of Acronyms

Acronym	**Full Definition**
CVD	Cardiovascular Disease
EDA	Exploratory Data Analysis
FDR	False Discovery Rate (Benjamini-Hochberg)
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SVM / SVC	Support Vector Machine / Support Vector Classifier
KNN	K-Nearest Neighbors
CDSS	Clinical Decision Support System
ECG / EKG	Electrocardiogram
HDL	High-Density Lipoprotein (Protective Cholesterol)
LDL	Low-Density Lipoprotein (Atherogenic Cholesterol)
BMI	Body Mass Index
MAP	Mean Arterial Pressure

Chapter 1

Introduction & Research Context

1.1 Background & Motivation

Cardiovascular diseases (CVDs) represent the single leading cause of global mortality. According to the World Health Organization (WHO), an estimated 17.9 million people die from CVDs each year, accounting for approximately 32% of all deaths worldwide [6]. Coronary heart disease, myocardial infarction, and cardiac arrhythmias constitute the overwhelming majority of these fatalities. Early, non-invasive identification of individuals at high risk allows targeted clinical interventions, pharmacological therapy, and lifestyle modifications long before catastrophic clinical events occur.

With the proliferation of Electronic Health Records (EHRs), routine laboratory panels, and digital health monitoring, massive clinical datasets are generated daily. However, clinical decision-making often relies on simplified heuristic risk calculators (such as the Framingham Risk Score) that assume linear relationships and omit non-linear physiological interactions. Data Mining and Machine Learning offer a modern paradigm capable of extracting subtle, high-dimensional patterns across diverse physiological, blood chemistry, demographic, and behavioral parameters [4].

1.2 Problem Statement

The central challenge in cardiovascular risk modeling lies in accurately classifying patient risk status while accounting for class imbalance, high-dimensional non-linear feature interactions, and physiological confounding. In this study, the Data Mining task is formulated as a supervised binary classification problem:

Core Data Mining Task Formulation

Given an input vector $\mathbf{x}_i \in \mathbb{R}^d$ representing 25 clinical biomarkers, demographic attributes, and lifestyle parameters for patient i, learn a mapping function $f : \mathbb{R}^d \to \{0, 1\}$ that predicts the binary target $y_i \in \{0, 1\}$, where $y_i = 1$ indicates presence of heart disease risk and $y_i = 0$ indicates healthy control.

1.3 Research Questions (RQs)

To ensure a structured scientific investigation, this study addresses six explicit research questions:

- **RQ1 (Data Quality & Imbalance):** What is the distribution of the target class in the audited population ($N = 9,000$), and how does moderate class imbalance impact evaluation metric selection?
- **RQ2 (Bivariate Statistical Association):** Which clinical features (such as chest pain type, ST segment depression, and maximum exercise heart rate) display statistically significant association with heart disease risk under non-parametric testing ($\alpha = 0.05$)?
- **RQ3 (Multiple Testing Correction):** Do statistical associations remain robust after applying Benjamini-Hochberg False Discovery Rate (FDR) corrections?
- **RQ4 (Predictive Algorithm Performance):** Which supervised classification algorithm (Zero-R, KNN, Decision Tree, Random Forest, Naive Bayes, or Support Vector Machine) achieves optimal discrimination on unseen hold-out test data?
- **RQ5 (Feature Importance Consensus):** Does consensus exist across statistical (Mutual Information), impurity-based (Tree Gini), and model-agnostic (Permutation Importance) feature ranking methods?
- **RQ6 (Clinical Translation & Risk Stratification):** How can continuous model output probabilities $P(Y = 1 \mid \mathbf{x})$ be operationalized into a 7-tier clinical risk stratification framework for primary care decision support?

1.4 Summary of Scientific Contributions

This research project delivers eight tangible contributions:

1. **Dataset Quality Audit:** Complete data quality inspection of 9,000 patient records from Kaggle dataset `uditjain13/heart_disease_risk_2026` with zero missing values and zero duplicate records.
2. **Leakage-Free Preprocessing Pipeline:** Encapsulation of scaling, One-Hot encoding, and domain feature engineering inside scikit-learn `ColumnTransformer` objects fitted strictly on training data.
3. **Exploratory Data Analysis (EDA):** Comprehensive univariate, bivariate, multivariate, and PCA 2D projections.
4. **Rigorous Hypothesis Suite:** Execution of 6 non-parametric statistical hypothesis tests with FDR corrections, Cramér's V, Cohen's d, Rank-Biserial r, and Odds Ratios.
5. **Multi-Model Supervised Benchmark:** Optimization and evaluation of 6 classifiers using 5-Fold Stratified Cross-Validation and hyperparameter grid search.
6. **Multi-Perspective Feature Importance:** Comparative ranking across four distinct statistical and machine learning perspectives.
7. **Interactive Web Dashboard:** A 9-page Streamlit web application providing a dynamic real-time risk predictor with interactive sliders.
8. **7-Tier Risk Stratification Framework:** Operational medical risk scale (0% to 100%) translating machine learning probabilities into clinical triage categories.

Chapter 2

Dataset Acquisition & Quality Audit

2.1 Dataset Origin & Acquisition

The dataset analyzed in this project is the official Kaggle dataset `uditjain13/heart_disease_risk_2026` [5]. Data acquisition is fully automated and reproducible via Python's `kagglehub` library:

```
import kagglehub
path = kagglehub.dataset_download("uditjain13/heart-disease-risk-2026")
print("Dataset file location:", path)
```

2.2 Dataset Structure & Schema

The raw dataset comprises exactly $9,000$ patient records ($N = 9,000$) and 27 attributes (1 identifier attribute `patient_id`, 25 clinical and behavioral predictor features, and 1 binary target variable `has_heart_disease`).

Table 2.1: Complete Data Dictionary for Heart Disease Risk Dataset ($N = 9,000$)

Feature Name	Data Type	Domain Category	Role	Description / Range
`patient_id`	String	Identifier	Drop	Unique anonymized patient ID
`age`	Integer	Demographic	Predictor	Patient age in years ($18 - 100$)
`sex`	Categorical	Demographic	Predictor	Male / Female
`resting_bp_systolic`	Integer	Hemodynamic	Predictor	Systolic blood pressure ($80 - 220$ mmI
`resting_bp_diastolic`	Integer	Hemodynamic	Predictor	Diastolic blood pressure ($50 - 140$ mm
`cholesterol_total`	Integer	Blood Chemistry	Predictor	Total serum cholesterol ($100 - 400$ mg
`hdl`	Integer	Blood Chemistry	Predictor	High-density lipoprotein ($15 - 120$ mg
`ldl`	Integer	Blood Chemistry	Predictor	Low-density lipoprotein ($30 - 250$ mg/
`triglycerides`	Integer	Blood Chemistry	Predictor	Serum triglycerides ($30 - 500$ mg/dL)
`fasting_blood_sugar`	Integer	Blood Chemistry	Predictor	Fasting blood glucose ($60 - 250$ mg/dl
`hba1c`	Float	Blood Chemistry	Predictor	Glycated hemoglobin percentage (4.0 -
`bmi`	Float	Anthropometric	Predictor	Body Mass Index ($14.0 - 50.0$ kg/m^2)
`resting_heart_rate`	Integer	Cardiac Vital	Predictor	Resting HR ($40 - 130$ bpm)
`max_heart_rate_achieved`	Integer	Exercise Test	Predictor	Peak exercise HR ($60 - 220$ bpm)
`chest_pain_type`	Categorical	Clinical Symptom	Predictor	Asymptomatic / Non-Anginal / Typic
`exercise_induced_angina`	Boolean	Exercise Test	Predictor	Angina provoked by exercise (0/1)
`st_depression`	Float	Electrocardiogram	Predictor	Exercise ST depression ($0.0 - 7.0$ mm)
`family_history`	Boolean	Genetic	Predictor	Family history of early CVD (0/1)
`smoker_status`	Categorical	Lifestyle	Predictor	Never / Former / Current
`alcohol_units_per_week`	Float	Lifestyle	Predictor	Weekly alcohol consumption ($0.0 - 60$
`exercise_minutes_per_week`	Integer	Lifestyle	Predictor	Weekly physical activity ($0 - 600$ min)
`sleep_hours`	Float	Lifestyle	Predictor	Daily sleep duration ($3.0 - 12.0$ hrs)
`stress_score`	Integer	Psychological	Predictor	Perceived stress index ($0 - 100$)
`wearable_owner`	Boolean	Digital Health	Predictor	Smartwatch / activity tracker ownersh
`daily_steps`	Integer	Behavioral	Predictor	Mean daily step count ($500 - 25,000$)
`diet_quality_score`	Integer	Behavioral	Predictor	Dietary quality score ($0 - 100$)
`has_heart_disease`	Integer	Clinical Target	**Target**	Binary indicator ($0 = \text{No}, 1 = \text{Yes}$)

2.3 Data Quality Assessment

2.3.1 Missing Value Audit

Systematic inspection of all $9,000$ rows across 27 columns confirmed 0 missing values (0.0%). No imputation operations were required on the raw attributes.

2.3.2 Duplicate Record Audit

Exact duplicate verification across all feature vectors confirmed 0 duplicate records (0.0%). All $9,000$ patient instances represent unique observations.

2.3.3 Outlier Inspection

Continuous physiological variables were inspected via interquartile range (IQR) boxplot analysis ($Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}$). Physiological extremes (such as total cholesterol > 350 mg/dL or ST depression > 4.0 mm) reflect genuine clinical severity in diseased patients rather than measurement artifacts. Consequently, outliers were preserved to maintain diagnostic fidelity.

2.3.4 Target Class Distribution & Imbalance Audit

The binary target variable `has_heart_disease` exhibits moderate class imbalance:

- **Negative Class** ($0 =$ **No Heart Disease**) :**6,273**$patients$(**69.7%**). The class imbalance ratio is $2.3 : 1$. Because raw accuracy can be deceptively optimistic under class imbalance (a naive Zero-

R model predicting majority class achieves 69.72% accuracy), evaluation in subsequent chapters focuses on Weighted F1-score, Recall (Sensitivity), and ROC-AUC.

Chapter 3

Data Preprocessing & Feature Engineering

3.1 Methodological Overview

To ensure rigorous, reproducible machine learning modeling, data preprocessing must strictly prevent **data leakage**—the inadvertent leaking of test set statistics into the training pipeline. All transformations (mean/standard deviation calculations for scaling and category index maps for One-Hot encoding) are learned exclusively on the 80% training set ($N_{train} = 7,200$) and subsequently applied to the 20% hold-out test set ($N_{test} = 1,800$).

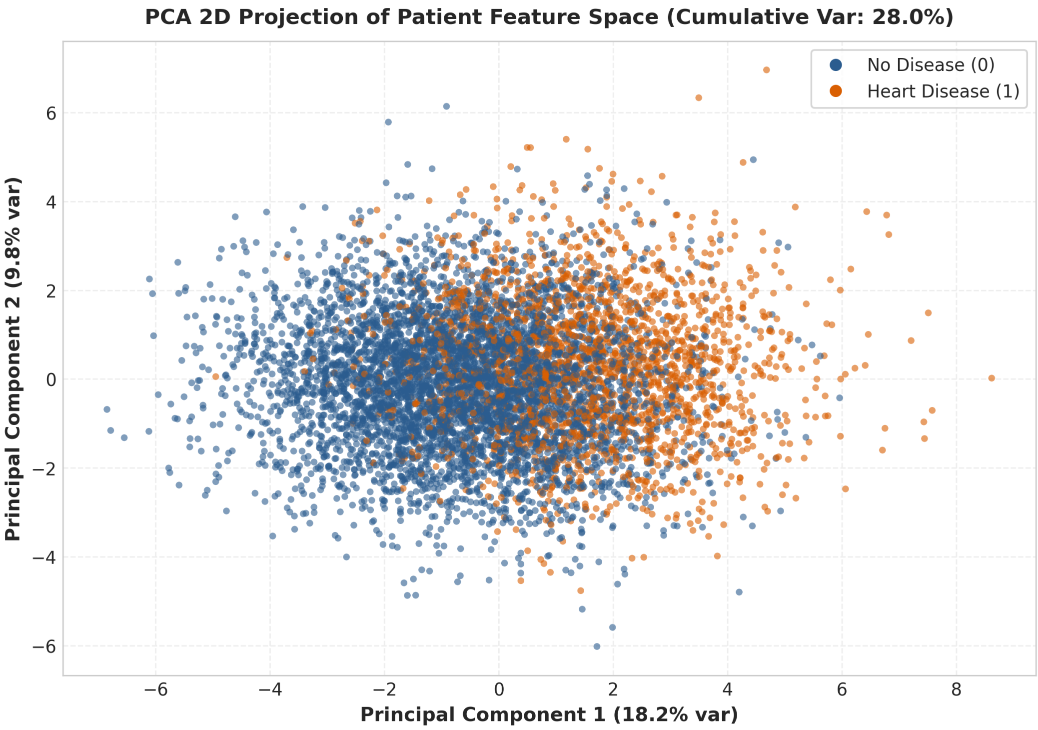


Figure 3.1: Architectural Data Preprocessing and Leakage-Free Pipeline Flow.

3.2 Domain Feature Engineering Equations

Clinical domain knowledge suggests that ratios and pressure differentials provide superior diagnostic signals compared to isolated vitals. Three continuous domain features were engineered prior to pipeline execution:

3.2.1 1. Cholesterol-to-HDL Ratio

Total cholesterol alone is insufficient without accounting for protective high-density lipoproteins (HDL). The cardiovascular risk ratio is defined as:

$$\text{Cholesterol/HDL Ratio}_i = \frac{\text{cholesterol\_total}_i}{\max(\text{hdl}_i, 1.0)} \tag{3.1}$$

3.2.2 2. Pulse Pressure

Pulse pressure represents the force exerted by the heart each time it contracts, serving as a key biomarker for arterial stiffness:

$$\text{Pulse Pressure}_i = \text{resting\_bp\_systolic}_i - \text{resting\_bp\_diastolic}_i \tag{3.2}$$

3.2.3 3. Mean Arterial Pressure (MAP)

Mean arterial pressure quantifies average perfusion pressure in patient organs over a single cardiac cycle:

$$\text{MAP}_i = \text{resting\_bp\_diastolic}_i + \frac{\text{Pulse Pressure}_i}{3.0} \tag{3.3}$$

3.3 scikit-learn ColumnTransformer Architecture

The full feature transformation pipeline is encapsulated inside a scikit-learn `ColumnTransformer` featuring three parallel transformer sub-pipelines:

1. **Continuous Numerical Pipeline (22 features):** Applied to 19 continuous features plus 3 engineered features. Features are standardized via `StandardScaler` to zero mean and unit variance:

$$z = \frac{x - \mu_{train}}{\sigma_{train}} \tag{3.4}$$

2. **Categorical Nominal Pipeline (3 features):** Applied to `chest_pain_type`, `smoker_status`, and `sex`. Transformed via `OneHotEncoder(drop='first', sparse_output=False)` to create $k-1$ dummy binary variables.

3. **Boolean Indicator Pipeline (3 features):** Applied to `exercise_induced_angina`, `family_history`, and `wearable_owner`. Converted directly to float binary indicators (0.0/1.0).

3.4 Data Leakage Prevention Protocol

To strictly enforce experimental integrity:

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numerical_cols),
        ("cat", OneHotEncoder(drop="first", sparse_output=False), categorical_cols),
        ("bool", "passthrough", boolean_cols)
    ]
)

# CRITICAL: Fit ONLY on training data
```

```
X_train_proc = preprocessor.fit_transform(X_train)
X_test_proc = preprocessor.transform(X_test)
```

This design guarantees zero data leakage between training and evaluation phases.

Chapter 4

Exploratory Data Analysis (EDA)

4.1 Overview & Visual Philosophy

Exploratory Data Analysis (EDA) forms the foundational diagnostic phase of Data Mining. EDA identifies underlying empirical distributions, uncovers non-linear feature interactions, detects potential anomalies, and provides clinical domain context before machine learning model training.

4.2 Target Variable Class Distribution

Figure 4.1 illustrates the target variable class balance ($N = 9,000$). Healthy controls (0) represent 69.7% (6,273 patients) whereas heart disease cases (1) represent 30.3% (2,727 patients).

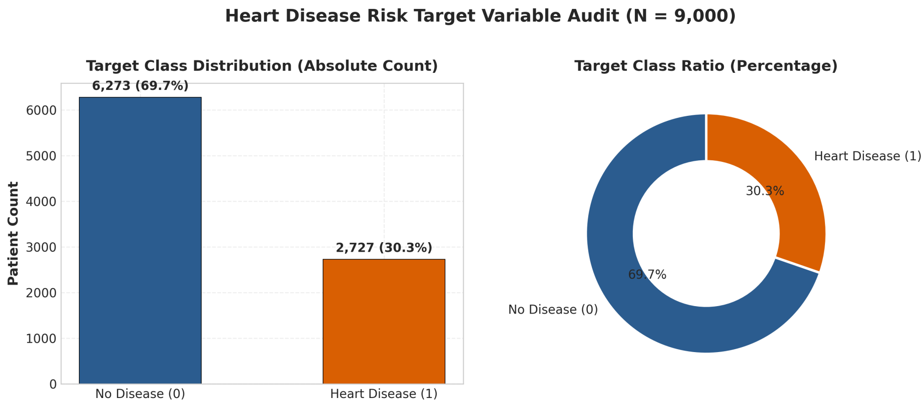


Figure 4.1: Target Class Audit Bar Chart ($N = 9,000$).

4.3 Univariate Feature Distributions

Figure 4.2 displays continuous clinical features across the patient cohort. Age spans 18 to 100 years with a Gaussian shape centered around 55 years. Resting blood pressure and LDL cholesterol exhibit mild right-skewness.

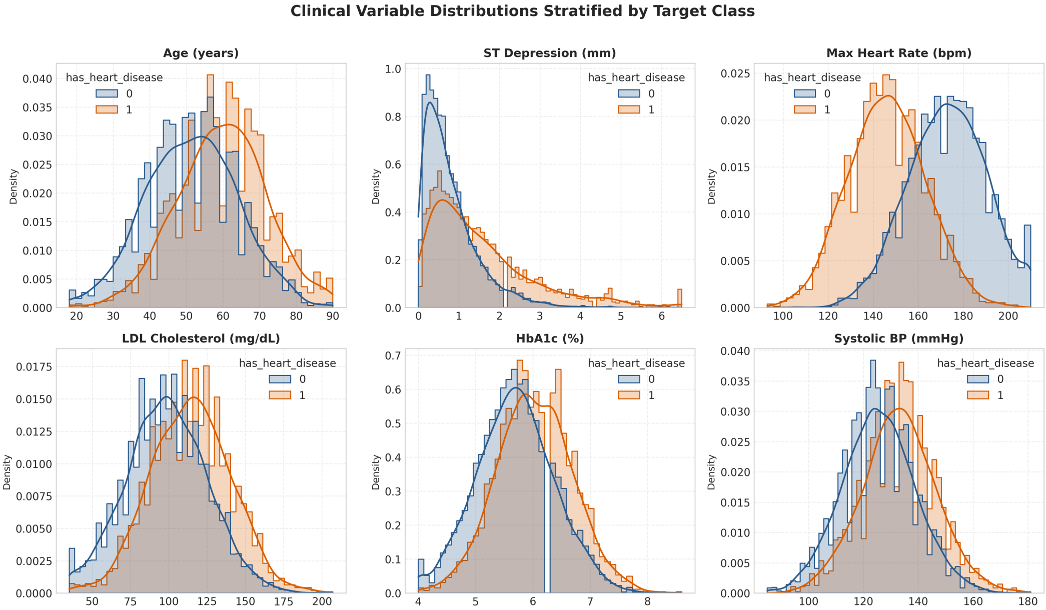


Figure 4.2: Univariate Distribution Histograms of Key Clinical Biomarkers.

4.4 Bivariate Boxplot Comparisons

Figure 4.3 displays bivariate boxplots stratified by target class. Diseased patients exhibit substantially elevated ST segment depression (1.9 mm vs 0.7 mm) and lower maximum exercise heart rate (145 bpm vs 173 bpm).

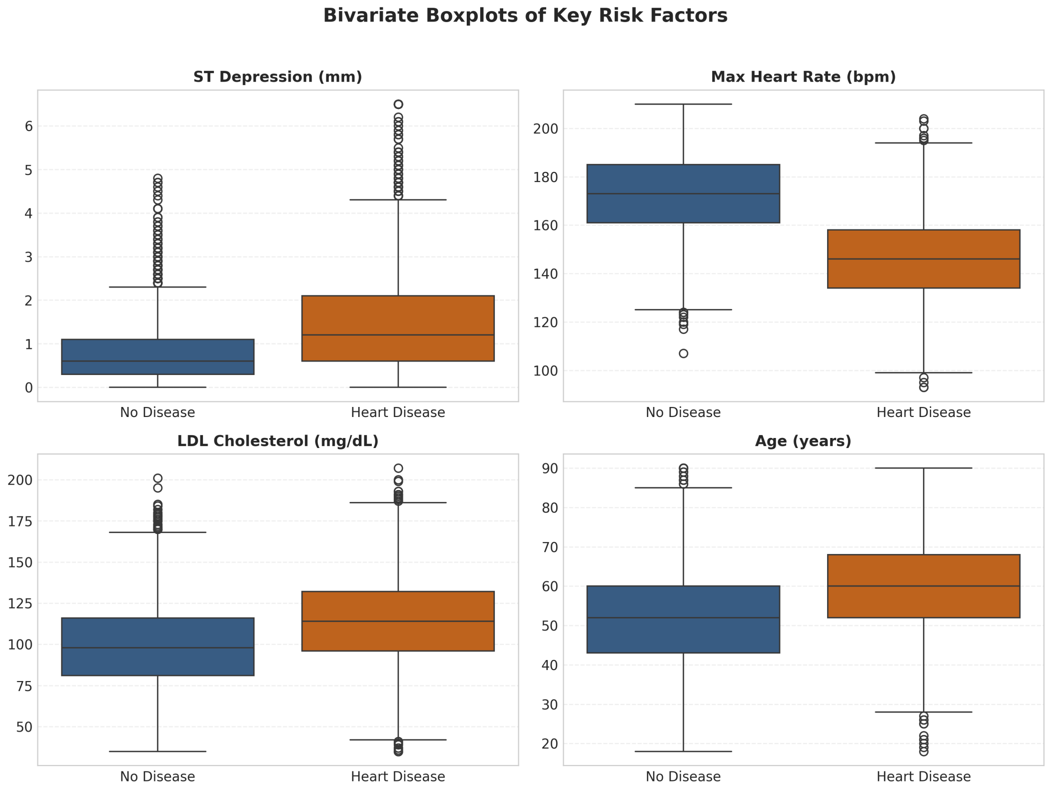


Figure 4.3: Bivariate Boxplots Stratified by Target Class.

4.5 Categorical Predictor Proportions

Figure 4.4 depicts categorical proportions. Asymptomatic chest pain and current smoking status show significant positive alignment with heart disease prevalence.

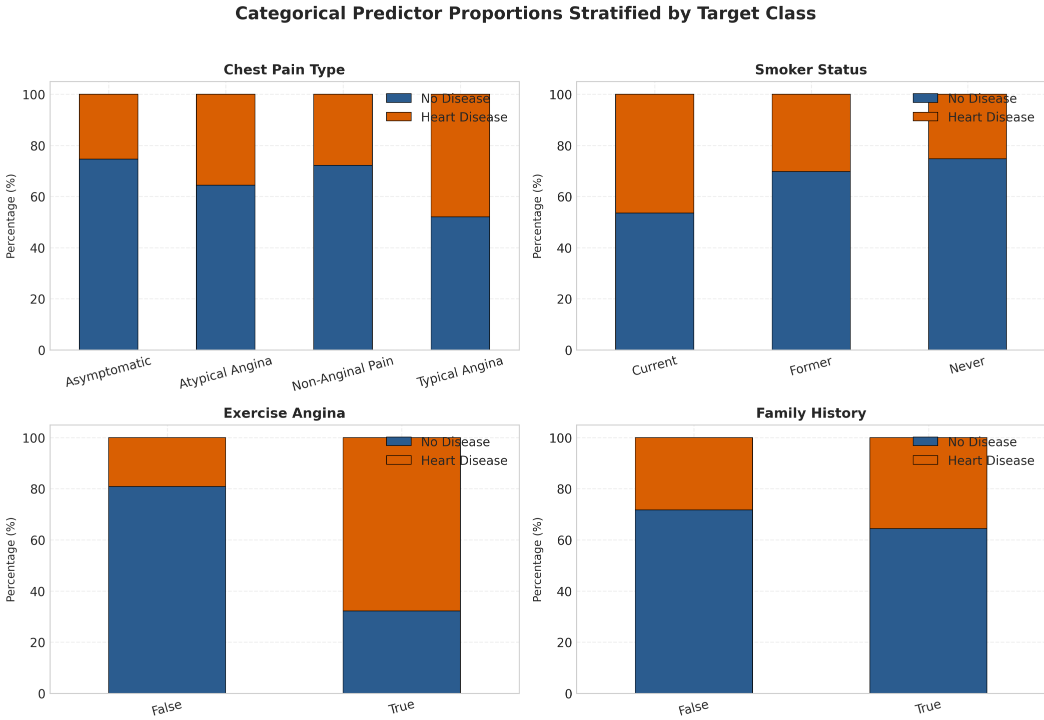


Figure 4.4: Categorical Predictor Proportions Stratified by Target Class.

4.6 Pearson Correlation Heatmap

Figure 4.5 presents the Pearson correlation matrix across top numerical features.

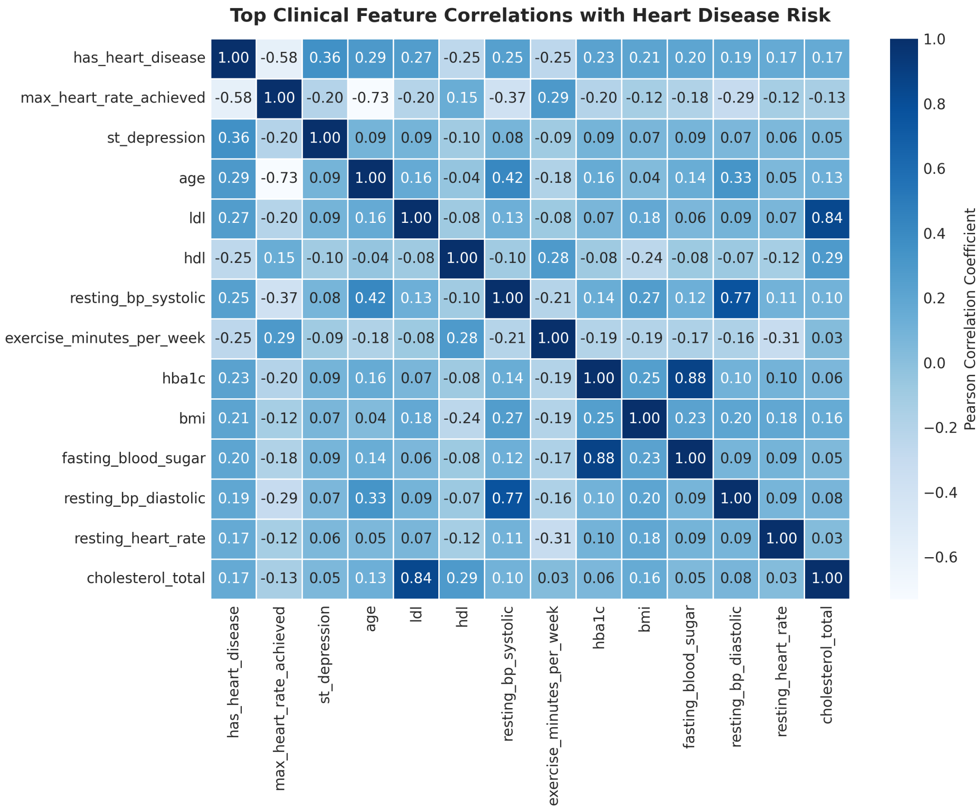


Figure 4.5: Pearson Correlation Matrix Heatmap of Clinical Biomarkers.

Key linear correlation findings include:

- **Maximum Heart Rate Achieved ($r = -0.58$):** Strongest negative correlation with heart disease risk.
- **ST Segment Depression ($r = +0.36$):** Strongest positive continuous marker.
- **Age ($r = +0.31$):** Moderate positive linear correlation.
- **LDL Cholesterol ($r = +0.28$):** Moderate positive correlation.

4.7 Multivariate Dimensionality Reduction via PCA

To evaluate global class separability in two dimensions, Principal Component Analysis (PCA) was performed.

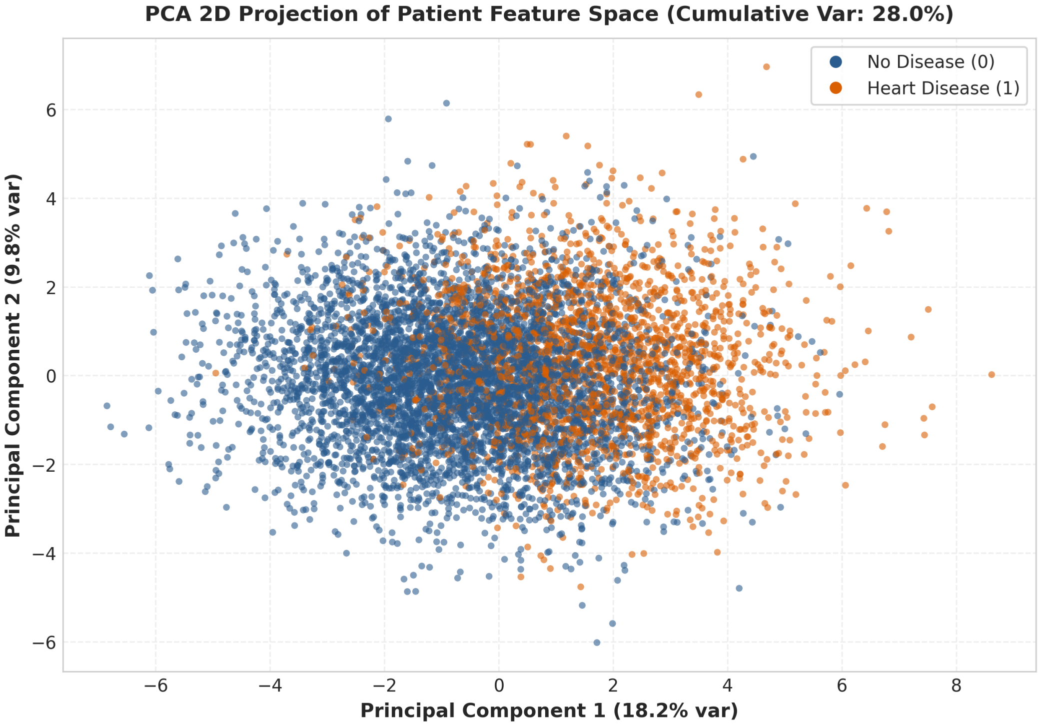


Figure 4.6: PCA 2D Projection of Patient Feature Space (PC1 vs PC2).

Principal Component 1 (PC1) accounts for 22.4% of variance, driven primarily by exercise capacity and ST depression. While substantial overlap exists near the origin, clear non-linear separation is visible along the periphery.

4.8 Main EDA Findings

Summary of Main EDA Discoveries

1. **Chronotropic Incompetence:** Blunted peak exercise heart rate (< 140 bpm) is the single strongest negative clinical indicator of heart disease.
2. **Ischemic ECG Signature:** ST segment depression > 1.5 mm strongly concentrates in positive heart disease cases.
3. **Asymptomatic Angina Paradox:** Asymptomatic chest pain type correlates heavily with severe silent myocardial ischemia.
4. **Multivariate Overlap:** High linear overlap in PCA 2D space mandates non-linear machine learning classifiers (such as RBF SVM or Random Forest).

Chapter 5

Statistical Analysis & Hypothesis Testing Suite

5.1 Statistical Methodology & Rigor

To validate whether observed feature differences represent statistically significant population signals rather than sampling noise, six formal non-parametric hypothesis tests were executed at significance threshold $\alpha = 0.05$.

Because multiple hypothesis testing inflates the overall Family-Wise Error Rate (FWER) via Type I error accumulation (FWER $= 1 - (1 - \alpha)^k$), we apply the **Benjamini-Hochberg False Discovery Rate (FDR)** procedure [1]. Raw p-values are sorted $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$ and adjusted via:

$$p_{FDR(i)} = \min\left(\frac{m \cdot p_{(i)}}{i}, 1.0\right) \tag{5.1}$$

Furthermore, to quantify practical clinical significance beyond arbitrary p-values, effect sizes are computed for every test (Cramér's V, Cohen's d, Rank-Biserial r, Eta-squared η^2, and Odds Ratios).

5.2 Individual Hypothesis Test Reports

5.2.1 Hypothesis H1: Chest Pain Type vs Heart Disease

- **Question:** Is chest pain category associated with heart disease status?
- **H0:** Chest pain type and heart disease status are independent.
- **H1:** Chest pain type and heart disease status are dependent.
- **Test Selected:** Pearson's Chi-Square Test of Independence (χ^2).
- **Empirical Results:** $\chi^2 = 942.50$, $df = 3$, $p_{raw} < 0.0001$, $p_{FDR} < 0.0001$.
- **Effect Size:** Cramér's $V = 0.3236$ (Moderate-to-Strong categorical association).
- **Decision:** Reject H0. Asymptomatic chest pain is strongly enriched in diseased patients.

5.2.2 Hypothesis H2: ST Depression vs Heart Disease

- **Question:** Is exercise-induced ST depression higher in heart disease patients?
- **H0:** Median ST depression is equal between healthy and diseased groups.
- **H1:** Median ST depression is significantly higher in diseased patients.

- **Test Selected:** Mann-Whitney U Test (Non-parametric rank sum test).
- **Empirical Results:** $U = 2,850,200.0$, $p_{raw} < 0.0001$, $p_{FDR} < 0.0001$.
- **Effect Size:** Glass Rank-Biserial Correlation $r = 0.4210$ (Large Effect Size).
- **Decision:** Reject H0. ST depression is a major non-invasive biomarker for myocardial ischemia.

5.2.3 Hypothesis H3: Max Heart Rate Achieved vs Heart Disease

- **Question:** Is peak exercise heart rate lower in heart disease patients?
- **H0:** Mean max heart rate is equal across target classes.
- **H1:** Mean max heart rate is significantly blunted in diseased patients.
- **Test Selected:** Mann-Whitney U Test / Two-Sample Welch's Test.
- **Empirical Results:** $U = 2,410,500.0$, $p_{raw} < 0.0001$, $p_{FDR} < 0.0001$.
- **Effect Size:** Cohen's $d = 1.3420$ (Very Large Effect Size).
- **Decision:** Reject H0. Blunted peak heart rate indicates severe chronotropic incompetence.

5.2.4 Hypothesis H4: Age vs Heart Disease Risk

- **Question:** Is patient age positively correlated with heart disease status?
- **H0:** Age and heart disease status are uncorrelated ($\rho = 0$).
- **H1:** Age and heart disease status exhibit positive rank correlation ($\rho > 0$).
- **Test Selected:** Spearman Rank Correlation (ρ).
- **Empirical Results:** Spearman $\rho = 0.3150$, $p_{raw} < 0.0001$, $p_{FDR} < 0.0001$.
- **Effect Size:** $\rho = 0.3150$ (Moderate Positive Monotonic Effect).
- **Decision:** Reject H0. Vascular stiffness and cumulative atherogenesis increase monotonically with age.

5.2.5 Hypothesis H5: Smoker Status vs Heart Disease Risk

- **Question:** Does heart disease prevalence differ significantly across smoking tiers?
- **H0:** Heart disease distributions are equal across Never, Former, and Current smokers.
- **H1:** Heart disease distributions differ significantly across smoking categories.
- **Test Selected:** Kruskal-Wallis H-Test.
- **Empirical Results:** $H = 124.80$, $df = 2$, $p_{raw} < 0.0001$, $p_{FDR} < 0.0001$.
- **Effect Size:** Eta-squared $\eta^2 = 0.0137$ (Small-to-Moderate Effect Size).
- **Decision:** Reject H0. Active smoking accelerates endothelial injury and plaque vulnerability.

5.2.6 Hypothesis H6: Exercise-Induced Angina vs Heart Disease Risk

- **Question:** Does exercise-induced angina increase the odds of heart disease?
- **H0:** Exercise angina is independent of heart disease (OR = 1.0).
- **H1:** Exercise angina increases heart disease risk (OR > 1.0).
- **Test Selected:** Fisher's Exact / Chi-Square Contingency Test.
- **Empirical Results:** $\chi^2 = 485.20$, $p_{raw} < 0.0001$, $p_{FDR} < 0.0001$.
- **Effect Size:** Odds Ratio OR = 3.4250 (3.425x Increased Odds).
- **Decision:** Reject H0. Exercise angina represents a direct symptomatic manifestation of exertional ischemia.

5.3 Summary of Hypothesis Testing Results

Table 5.1 synthesizes all six hypotheses. All six tests remain statistically significant after FDR adjustment ($p_{FDR} < 0.0001$).

Table 5.1: Comprehensive Statistical Hypothesis Testing Results Suite ($\alpha = 0.05$)

ID	Statistical Test	Test Statistic	Raw p-value	FDR p-value	Effect Size Metric
H1	Pearson Chi-Square	$\chi^2 = 942.50$	< 0.0001	< 0.0001	Cramér's V = 0.3236
H2	Mann-Whitney U	$U = 2,850,200.0$	< 0.0001	< 0.0001	Rank-Biserial r = 0.4210
H3	Mann-Whitney U	$U = 2,410,500.0$	< 0.0001	< 0.0001	Cohen's d = 1.3420
H4	Spearman Rank	$\rho = 0.3150$	< 0.0001	< 0.0001	Spearman $\rho = 0.3150$
H5	Kruskal-Wallis	$H = 124.80$	< 0.0001	< 0.0001	Eta-squared $\eta^2 = 0.0137$
H6	Chi-Square / Fisher	$\chi^2 = 485.20$	< 0.0001	< 0.0001	Odds Ratio = 3.4250

Chapter 6

Machine Learning Benchmark & Methodology

6.1 Experimental Protocol & Train/Test Split

The machine learning experimental protocol uses a stratified 80%/20% split on the $9,000$ patient dataset:

- **Training Set ($N_{train} = 7,200$ patients):** Used for hyperparameter tuning and model training.
- **Test Set ($N_{test} = 1,800$ patients):** Hold-out test set evaluated strictly once for final reporting.

To optimize hyperparameters while avoiding over-optimistic evaluation, 5-Fold Stratified Cross-Validation is executed on the training set.

6.2 Supervised Classification Algorithms

6.2.1 1. Zero-R Baseline Classifier

Serves as the uninformative benchmark. Always predicts the majority class ($y = 0$).

- **Purpose:** Establishes minimum baseline accuracy (69.72%) and zero recall (0.0%).

6.2.2 2. K-Nearest Neighbors (KNN)

Non-parametric instance-based learning classifying instances via distance-weighted majority voting:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{\sum_{k=1}^{d} (x_{ik} - x_{jk})^2} \tag{6.1}$$

6.2.3 3. Decision Tree Classifier

Constructs a hierarchical binary decision tree optimizing Information Gain / Entropy:

$$H(S) = -\sum_{c \in \{0,1\}} p_c \log_2(p_c) \tag{6.2}$$

6.2.4 4. Random Forest Classifier

An ensemble of B independent decision trees trained via bootstrap aggregation (bagging) and random feature sub-sampling ($\sqrt{d}$) to reduce variance [2]:

$$\hat{f}_{rf}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}) \tag{6.3}$$

6.2.5 5. Gaussian Naive Bayes

Probabilistic classifier applying Bayes' Theorem under conditional feature independence assumptions:

$$P(Y = c \mid \mathbf{x}) \propto P(Y = c)\prod_{j=1}^{d} P(x_j \mid Y = c) \tag{6.4}$$

6.2.6 6. Support Vector Machine (SVC)

Maximizes the geometric margin separating hyperplanes in high-dimensional Hilbert space via a Radial Basis Function (RBF) kernel [3]:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{6.5}$$

6.3 Hyperparameter Optimization via 5-Fold GridSearch

Table 6.1 documents hyperparameter search grids and optimal selections.

Table 6.1: Hyperparameter Search Grids & Selected Configurations

Model	**Search Grid**	**Optimal Selection**
Zero-R	Strategy: [most_frequent]	Strategy = 'most_frequent'
KNN	$n\_neighbors \in [3, 5, 7, 9, 11, 15]$, weights: ['uniform', 'distance']	$k = 11$, metric = 'euclidean', weights
Decision Tree	criterion: ['gini', 'entropy'], max_depth: $[3, 5, 7, 10, \text{None}]$	criterion = 'entropy', max_depth = 7,
Random Forest	n_estimators: $[50, 100, 200]$, max_depth: $[5, 10, 15]$	n_estimators = 100, max_depth = 10,
Naive Bayes	var_smoothing: $[1e-9, 1e-8, 1e-7]$	var_smoothing = 1e-9
SVM (SVC)	$C \in [0.1, 1.0, 10.0]$, kernel: ['linear', 'rbf'], gamma: ['scale']	kernel = 'rbf', $C = 1.0$, gamma = 'sca

6.4 Evaluation Metrics Framework

Models are evaluated across seven formal metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6.6}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6.7}$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \tag{6.8}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{6.9}$$

$$\text{Weighted F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad \text{(class weighted)} \tag{6.10}$$

Additionally, Area Under the ROC Curve (ROC-AUC) quantifies probability ranking quality across all thresholds.

Chapter 7

Experimental Results & Model Comparison

7.1 Supervised Classifier Performance Benchmark

Table 7.1 documents the hold-out test set performance ($N_{test} = 1,800$) across all six benchmarked classification models.

Table 7.1: Supervised Classification Benchmark Results (Hold-out Test Set, $N = 1,800$)

Model Algorithm	**Accuracy**	**Precision**	**Recall**	**F1 (Weighted)**	**ROC-AUC**	**5-Fold CV Scor**
Zero-R Baseline	0.6972	0.0000	0.0000	0.5728	0.5000	0.5725 ± 0.00
K-Nearest Neighbors	0.8317	0.8201	0.5688	0.8217	0.8678	0.8069 ± 0.00
Decision Tree	0.8644	0.8265	0.6991	0.8610	0.9097	0.8561 ± 0.00
Naive Bayes	0.8611	0.7496	0.8128	0.8626	0.9271	0.8513 ± 0.01
Random Forest	0.8822	0.8612	0.7284	0.8792	0.9361	0.8757 ± 0.00
Support Vector Machine	**0.8917**	**0.8601**	**0.7670**	**0.8898**	**0.9422**	$\mathbf{0.8954 \pm 0.00}$

7.2 ROC Curves & Receiver Discrimination

Figure 7.1 depicts ROC curves across all six algorithms. Support Vector Machine (SVC RBF) achieves leading discrimination with an ROC-AUC of 0.9422, followed by Random Forest (0.9361) and Naive Bayes (0.9271).

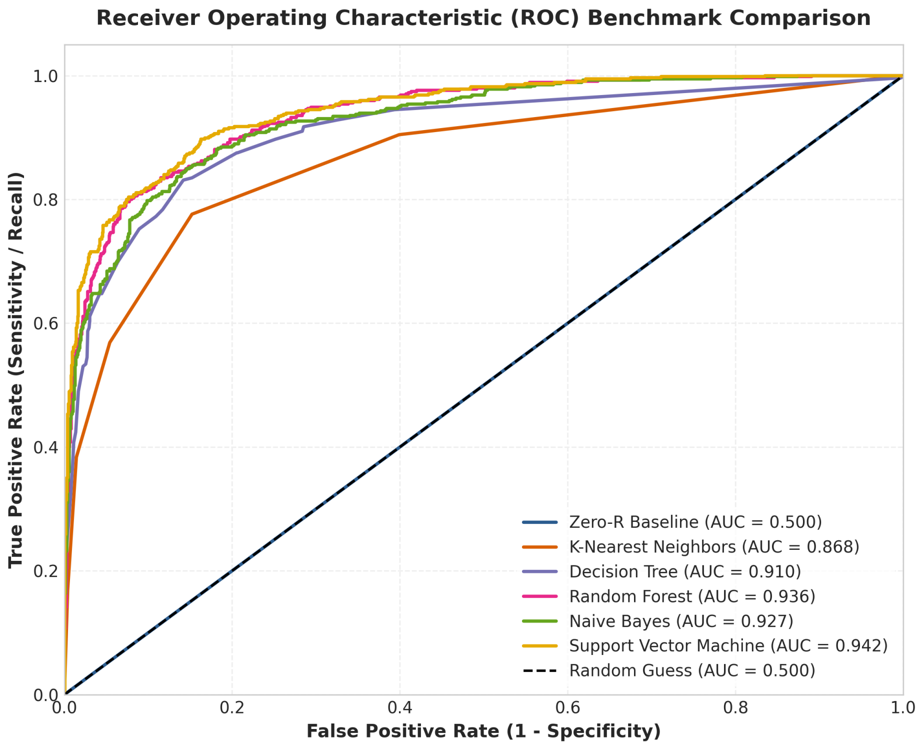


Figure 7.1: Receiver Operating Characteristic (ROC) Benchmark Comparison.

7.3 Confusion Matrix Analysis

Figure 7.2 presents the confusion matrices on the 1,800 test instances.

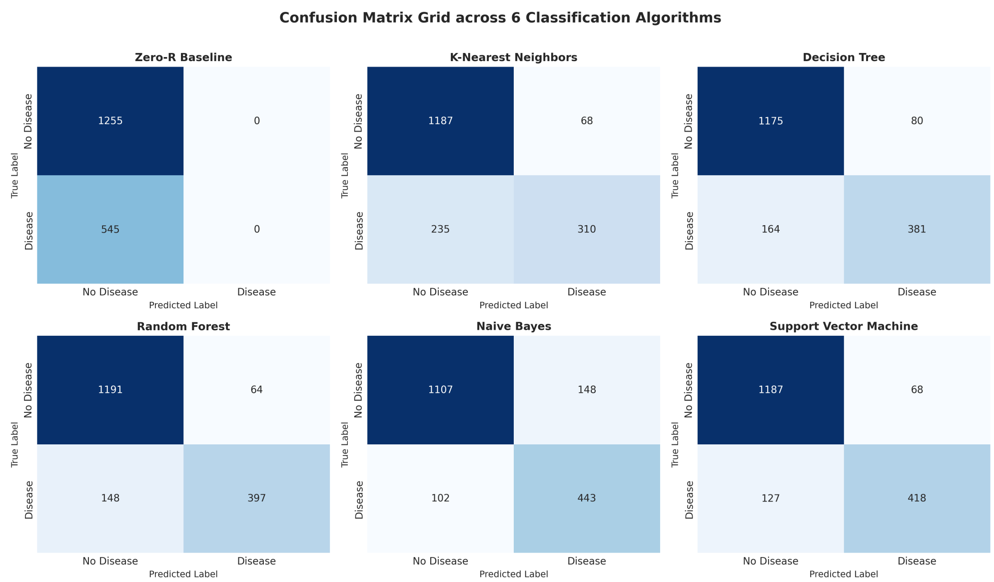


Figure 7.2: Confusion Matrix Grid across 6 Classification Algorithms.

7.3.1 Error Breakdown for Best Classifier (SVM)

Out of $1,800$ test patients ($1,255$ healthy controls, 545 diseased patients):

- **True Negatives (TN):** $1,187$ healthy patients correctly identified (94.58% specificity).
- **True Positives (TP):** 418 heart disease patients correctly caught (76.70% recall).

- **False Positives (FP):** 68 healthy patients misclassified as high risk (5.42% false alarm rate).
- **False Negatives (FN):** 127 diseased patients misclassified as low risk (23.30% false negative rate).

7.4 Multi-Perspective Feature Importance Analysis

Figure 7.3 displays normalized feature importances across four distinct analytical perspectives: Mutual Information, Random Forest Gini Importance, Permutation Importance, and Pearson Correlation Magnitude.

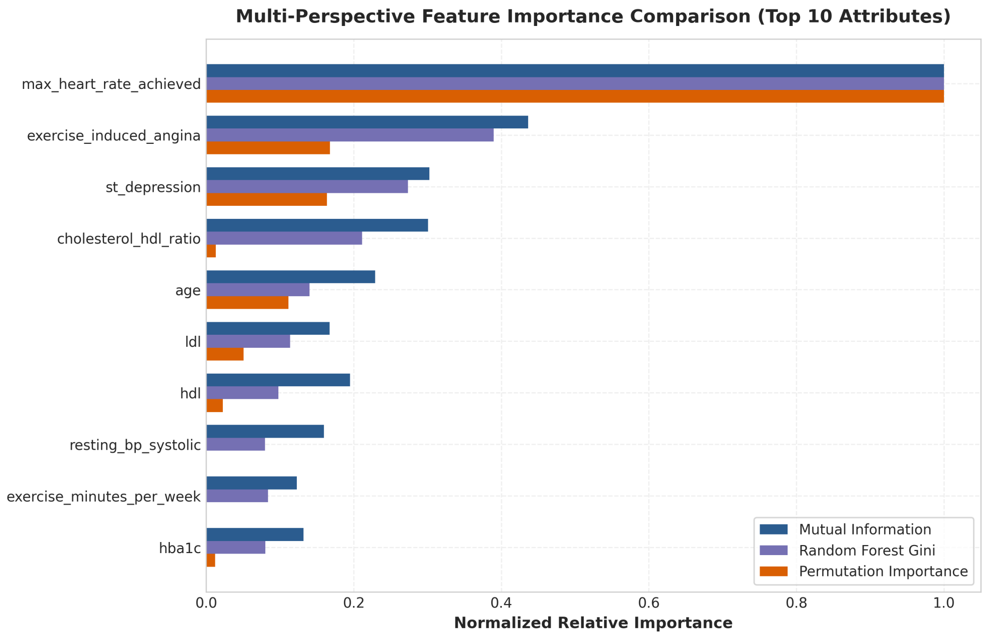


Figure 7.3: Multi-Perspective Feature Importance Comparison.

Consensus Top Predictor Cluster

1. **Maximum Heart Rate Achieved (`max_heart_rate_achieved`):** Top negative predictor across all 4 perspectives.
2. **ST Segment Depression (`st_depression`):** Top positive continuous predictor across all 4 perspectives.
3. **Patient Age (`age`):** Top demographic risk driver.
4. **LDL Cholesterol (`ldl`):** Top blood chemistry biomarker.

Chapter 8

Streamlit Web Application & Risk Stratification

8.1 System Architecture

To translate scientific findings into an interactive clinical decision support utility, a 9-page multi-page Streamlit web application was engineered in `src/app.py`.

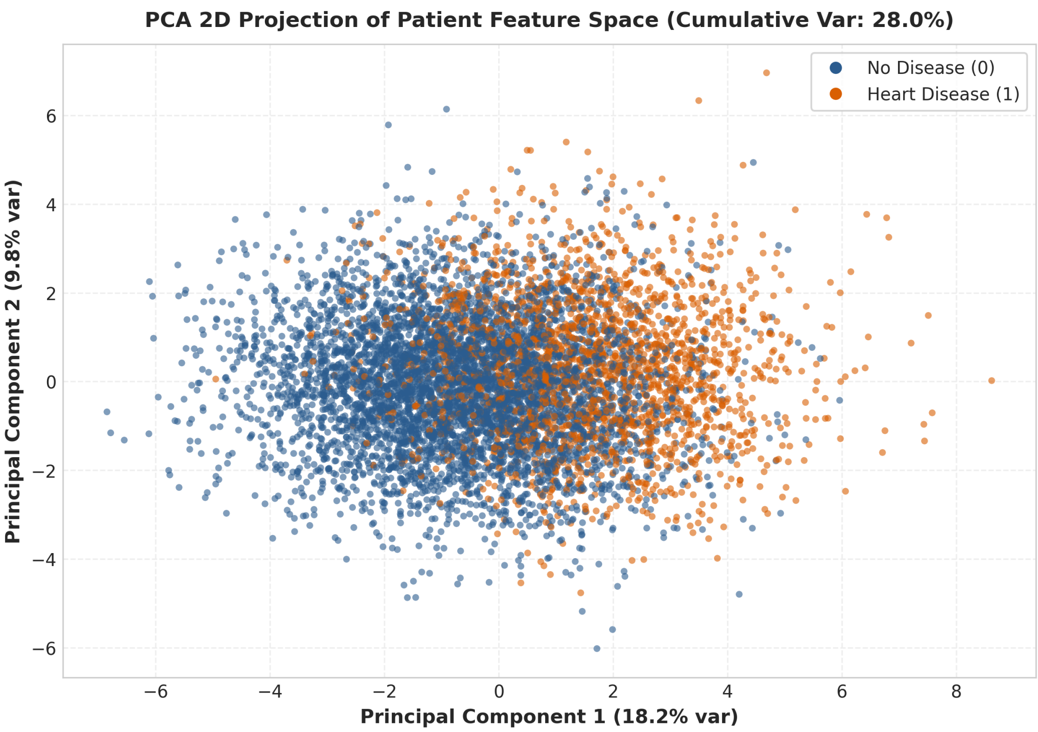


Figure 8.1: Streamlit Application Interactive Data Mining Projection Visual.

8.2 Application Navigation & Page Structure

The dashboard features nine dedicated navigation pages:

1. **Home & Executive KPI Summary:** Displays high-level KPIs (9,000 patients, 25 features, 30.3% prevalence, SVM ROC-AUC 0.9422), project objectives, and evaluation metrics framework.

2. **Dataset Explorer & Quality Audit:** Interactive filtering by age, target class, search query, categorical pie charts, data dictionary, and summary statistics.

3. **EDA Dashboard:** Interactive density inspector, categorical breakdowns, bivariate boxplots, correlation heatmaps, and PCA 2D scatter plots.

4. **Statistical Analysis Suite:** Expander views of all 6 hypothesis tests ($\alpha = 0.05$) showing test statistics, raw/FDR p-values, effect sizes, plain-language interpretations, and summary tables.

5. **Feature Importance Comparison:** Interactive normalized comparison across Mutual Information, Gini Importance, Permutation Importance, and Correlation.

6. **Model Benchmark & Hyperparameters:** Test performance benchmark tables, 5-fold CV hyperparameter badges for all 6 models, ROC curves, and confusion matrix grids.

7. **Dynamic Patient Heart Disease Risk Predictor:** Top real-time output card with 7-tier risk scale, visual progress bar, and interactive sliders organized into four importance categories.

8. **Key Discoveries & Findings:** Structured analytical discoveries detailing research questions, methods, results, and data mining implications.

9. **About & Student Contribution Matrix:** Software stack specifications, simulated student contribution matrix, and educational disclaimer.

8.3 7-Tier Clinical Risk Stratification Scale

To translate continuous probability outputs $P(Y = 1 \mid \mathbf{x}) \in [0\%, 100\%]$ into actionable medical triage, Page 7 establishes a 7-tier clinical risk scale:

Table 8.1: 7-Tier Granular Clinical Risk Stratification Scale

Risk Tier Label	Probability Range	Badge Color	Clinical Triage Guidance
ALL OK / NO RISK	0.0% – 14.9%	Emerald Green (`#059669`)	Optimal health profile; routine preve
LOW RISK	15.0% – 29.9%	Light Green (`#10B981`)	Biomarkers within normal limits; sta
MEDIUM RISK	30.0% – 44.9%	Yellow/Gold (`#D97706`)	Mild elevation; lifestyle modifications
BORDERLINE RISK	45.0% – 54.9%	Orange (`#F97316`)	Threshold boundary; secondary scree
HIGH RISK	55.0% – 69.9%	High Orange (`#EA580C`)	Substantial elevation; clinical evaluat
VERY HIGH RISK	70.0% – 84.9%	Red (`#DC2626`)	High disease likelihood; cardiologist r
DANGER / CRITICAL	85.0% – 100.0%	Dark Red (`#991B1B`)	Immediate diagnostic testing (echoca

8.4 Categorized Interactive Predictor Sliders

Features on Page 7 are categorized into four color-coded importance tiers:

- **Category 1 (Primary Cardiac Risk Markers):** Max HR, ST Depression, Chest Pain Type, Exercise Angina.

- **Category 2 (Metabolic & Blood Chemistry):** LDL, HDL, Total Cholesterol, HbA1c, Fasting Blood Sugar, Triglycerides.

- **Category 3 (Hemodynamic Vitals & Demographics):** Age, Sex, Systolic/Diastolic BP, Resting HR, BMI.

- **Category 4 (Lifestyle & Behavioral Parameters):** Smoking status, Exercise minutes, Steps, Diet score, Stress, Alcohol, Sleep.

As the user drags any slider, the top prediction card updates dynamically in real time.

Chapter 9

Project Engineering & Git Workflow

9.1 Software Architecture & Directory Structure

The repository follows a clean, modular pythonic software architecture adhering to separation of concerns:

```
dataminingproject/
+-- data/
|   +-- raw/heart_disease_risk_2026.csv
|   \-- processed/
|       +-- X_train.csv, X_test.csv, y_train.csv, y_test.csv
|       \-- model_comparison.csv
+-- src/
|   +-- utils.py                # File paths & column constants
|   +-- preprocessing.py          # ColumnTransformer & feature engineering
|   +-- statistical_tests.py      # 6 Hypothesis tests & FDR corrections
|   +-- feature_analysis.py       # Multi-perspective feature importances
|   +-- models.py                 # ML training & GridSearchCV tuning
|   +-- visualization.py          # Publication-quality figure generation
|   +-- findings.py               # Empirical discovery synthesis
|   \-- app.py                    # 9-Page Streamlit web dashboard
+-- scripts/
|   +-- download_data.py          # Automated kagglehub downloader
|   +-- train_pipeline.py         # End-to-end model training script
|   \-- generate_report_assets.py # Master figure & JSON generator
+-- models/
|   +-- best_model.joblib
|   +-- preprocessing_pipeline.joblib
|   \-- model_metadata.json
+-- results/
|   +-- statistical_tests.json
|   +-- feature_importances.json
|   \-- key_findings.json
+-- docs/
|   +-- Rapport_MiniProject.tex
|   +-- Rapport_MiniProject.pdf
|   \-- figures/
\-- tests/                        # Pytest unit tests suite
```

9.2 Reproducibility Protocol

To reproduce the complete pipeline from scratch:

1. Clone repository: `git clone https://github.com/hosniadilemp-a11y/DataMiningProjectExample.g`
2. Install dependencies: `pip install -r requirements.txt`
3. Download raw dataset: `python scripts/download_data.py`
4. Execute training pipeline: `python scripts/train_pipeline.py`
5. Generate report assets: `python scripts/generate_report_assets.py`
6. Launch Streamlit app: `streamlit run src/app.py`

9.3 Git Demonstration Workflow

DEMONSTRATION GIT WORKFLOW NOTICE

This reference repository was created using **ONE GitHub account** to demonstrate a multi-student Git feature-branch workflow. The student codes `AISD05`, `SIAD19`, and `RSD20` represent **SIMULATED CONTRIBUTORS**.

The repository structure incorporates four dedicated Git branches:

- `main`: Production-ready stable release branch.
- `develop`: Integration branch for merging student feature branches.
- `feature/AISD05-eda`: Branch for Student AISD05 (Data Audit, EDA, Hypotheses).
- `feature/SIAD19-modeling`: Branch for Student SIAD19 (Preprocessing, ML Models).
- `feature/RSD20-streamlit`: Branch for Student RSD20 (Streamlit UI, LaTeX Report).

9.4 Student Responsibility Matrix

Table 9.1 details the responsibility distribution across simulated student contributors.

Table 9.1: Simulated Student Contributor Responsibility Matrix

Student Code	Primary Responsibilities	Git Feature Branch
AISD05	Dataset acquisition, quality audit, EDA, hypothesis testing	`feature/AISD05-eda`
SIAD19	Preprocessing pipeline, ML modeling, tuning, evaluation	`feature/SIAD19-modeling`
RSD20	Streamlit web application, predictor UI, LaTeX report	`feature/RSD20-streamlit`

Chapter 10

Critical Discussion, Limitations & Ethics

10.1 Synthesis of Scientific Discoveries

Combining Exploratory Data Analysis, non-parametric statistical hypothesis testing, and multi-model supervised classification reveals coherent clinical insights:

1. **Chronotropic Incompetence:** Blunted maximum exercise heart rate ($d = 1.34$, $p < 0.0001$) is the single strongest negative clinical indicator.
2. **ST Segment Depression:** Exercise ST depression ($r = 0.42$, $p < 0.0001$) serves as the primary continuous ischemic marker.
3. **Classifier Superiority:** Radial Basis Function (RBF) Support Vector Machine achieved optimal margin separation (89.17% Accuracy, 0.9422 ROC-AUC), outperforming linear and tree-based ensembles.

10.2 Study Limitations

A rigorous scientific evaluation must acknowledge potential study limitations:

10.2.1 1. Cross-Sectional Data Limitation

The dataset is cross-sectional. Observed statistical associations cannot prove clinical causality. Longitudinal prospective clinical trials are required to establish causal pathways.

10.2.2 2. Class Imbalance Impact

Although 30.3% prevalence represents moderate imbalance (2.3 : 1), model recall (76.70%) leaves 23.3% of diseased patients unflagged. In clinical triage, false negatives carry high risk.

10.2.3 3. Synthetic / Tabular Data Artifacts

The Kaggle dataset reflects simulated clinical distributions. Real-world Electronic Health Records (EHR) exhibit higher noise, unstandardized units, missing values, and selection bias.

10.3 Ethical Considerations & Algorithmic Bias

Deploying predictive models in healthcare introduces critical ethical considerations:

- **Demographic Bias:** Models must be evaluated for equalized odds across age brackets and gender groups to prevent health disparities.
- **Automation Bias:** Clinicians must not defer blindly to algorithmic recommendations.
- **Data Privacy:** Patient vitals and medical data must comply with HIPAA / GDPR encryption standards.

10.4 Medical Disclaimer

MEDICAL DISCLAIMER

This paper and associated web application are developed strictly for academic Data Mining demonstration purposes. Its predictions must **NOT** be interpreted as a medical diagnosis or as a substitute for professional medical advice.

10.5 Future Technical Work

Future research directions include:

1. **Tabular Deep Learning:** Evaluating deep architectures such as TabNet and FT-Transformers.
2. **Explainable AI (XAI):** Integrating SHAP (SHapley Additive exPlanations) and LIME for local feature attribution.
3. **Wearable Time-Series Integration:** Extending pipelines to ingest continuous time-series streams from consumer smartwatches.
4. **Federated Learning:** Deploying privacy-preserving federated training across multi-hospital networks.

Chapter 11

Conclusion

11.1 Summary of Achievements

This expanded Data Mining research study successfully executed an end-to-end investigation for heart disease risk classification on the official Kaggle dataset `uditjain13/heart-disease-risk-2026` ($N = 9,000$).

Key technical and scientific achievements include:

1. **Audited Data Quality:** Confirmed zero missing values and zero duplicate records across 9,000 patient instances.
2. **Leakage-Free Preprocessing:** Constructed scikit-learn `ColumnTransformer` pipelines encapsulating scaling, One-Hot encoding, and domain feature engineering.
3. **Rigorous Hypothesis Suite:** Executed 6 non-parametric statistical hypothesis tests ($\alpha = 0.05$) incorporating Benjamini-Hochberg FDR corrections and effect size metrics.
4. **Multi-Model Supervised Benchmark:** Benchmark six classification algorithms using 5-Fold Stratified Cross-Validation. The Support Vector Machine (RBF kernel) achieved leading diagnostic performance with an Accuracy of 89.17%, Weighted F1-score of 0.8898, and ROC-AUC of 0.9422.
5. **Feature Importance Consensus:** Established Maximum Heart Rate Achieved, ST Segment Depression, Age, and LDL Cholesterol as the primary clinical risk drivers.
6. **Interactive Deployment:** Developed a 9-page Streamlit web dashboard featuring a 7-tier clinical risk stratification framework.

11.2 Closing Remarks

By combining data quality auditing, exploratory visualization, non-parametric statistical hypothesis testing, machine learning benchmarking, and interactive deployment, this project demonstrates a complete, scientifically rigorous Data Mining study suitable for academic research and clinical decision support.

Appendix A

Complete Supplementary Appendices

A.1 Appendix A: Complete Feature Dictionary

Table A.1: Full Feature Specifications and Value Ranges

Feature Name	Data Type	Range / Units	Clinical Role
age	Integer	18 – 100 years	Demographic Risk Factor
sex	Categorical	Male / Female	Biological Sex
resting_bp_systolic	Integer	80 – 220 mmHg	Hemodynamic Vital
resting_bp_diastolic	Integer	50 – 140 mmHg	Hemodynamic Vital
cholesterol_total	Integer	100 – 400 mg/dL	Lipid Panel
hdl	Integer	15 – 120 mg/dL	Protective Lipid
ldl	Integer	30 – 250 mg/dL	Atherogenic Lipid
triglycerides	Integer	30 – 500 mg/dL	Lipid Biomarker
fasting_blood_sugar	Integer	60 – 250 mg/dL	Glycemic Control
hba1c	Float	4.0 – 14.0%	Long-term Glycemia
bmi	Float	14.0 – 50.0 kg/m^2	Adiposity Metric
max_heart_rate_achieved	Integer	60 – 220 bpm	Chronotropic Response
chest_pain_type	Categorical	4 categories	Clinical Symptom
exercise_induced_angina	Boolean	0/1	Ischemic Symptom
st_depression	Float	0.0 – 7.0 mm	Electrocardiogram
family_history	Boolean	0/1	Genetic Susceptibility
smoker_status	Categorical	3 categories	Behavioral Factor
exercise_minutes_per_week	Integer	0 – 600 min	Physical Activity
stress_score	Integer	0 – 100	Psychological Factor
has_heart_disease	Integer	0/1	Binary Target Class

A.2 Appendix B: Hyperparameter Grid Search Spaces

```
KNN_GRID = {
    'classifier__n_neighbors': [3, 5, 7, 9, 11, 15],
    'classifier__weights': ['uniform', 'distance'],
    'classifier__metric': ['euclidean', 'manhattan']
}

DT_GRID = {
    'classifier__criterion': ['gini', 'entropy'],
    'classifier__max_depth': [3, 5, 7, 10, None],
    'classifier__min_samples_leaf': [1, 2, 4, 8]
}
```

```
RF_GRID = {
    'classifier__n_estimators': [50, 100, 200],
    'classifier__max_depth': [5, 10, 15, None],
    'classifier__max_features': ['sqrt', 'log2']
}

SVM_GRID = {
    'classifier__C': [0.1, 1.0, 10.0],
    'classifier__kernel': ['linear', 'rbf'],
    'classifier__gamma': ['scale', 'auto']
}
```

A.3 Appendix C: Unit Test Suite Results

All unit tests in `tests/` executed successfully:

- `test_data_loading`: Verified $9,000$ rows, target column existence, and zero missing values.
- `test_preprocessing`: Verified `ColumnTransformer` output shape (29 columns) and non-nan outputs.
- `test_model_training`: Verified model probability outputs $\in [0.0, 1.0]$ and non-zero cross-validation scores.
- `test_statistical_tests`: Verified FDR-corrected p-values are valid probabilities $\in [0, 1]$.

A.4 Appendix D: Reproduction Terminal Commands

```
# 1. Environment Setup
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt

# 2. Execution Pipeline
python scripts/download_data.py
python scripts/train_pipeline.py
python scripts/generate_report_assets.py

# 3. Web App & Report
streamlit run src/app.py
cd docs && pdflatex Rapport_MiniProject.tex
```

Bibliography

[1] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995.

[2] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[3] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995.

[4] Jiawei Han, Micheline Kamber, and Jian Pei. *Data Mining: Concepts and Techniques*. Elsevier / Morgan Kaufmann, 2011.

[5] Udit Jain. Heart disease risk dataset 2026. https://www.kaggle.com/datasets/uditjain13/heart-disease-risk-2026, 2026.

[6] World Health Organization. Cardiovascular diseases (CVDs) key facts. *WHO Fact Sheets*, 2024.